

INTRODUCTION

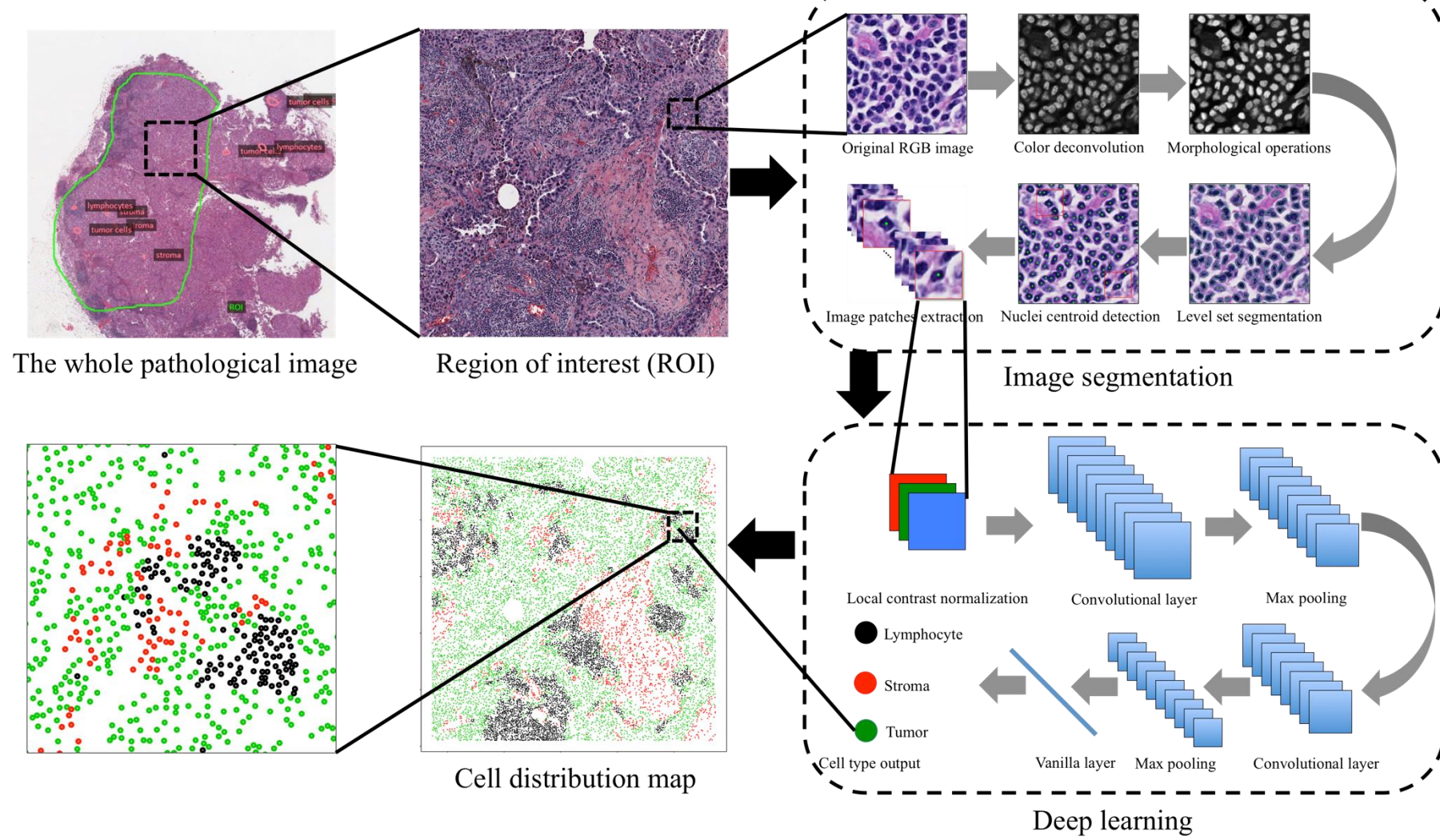
■ Motivation

- To summarize spatial cell-cell interactions in cancer tissue images using various spatial summary statistics.
- Studies have shown that cell-cell interactions in Tumor Microenvironment play key role in tumor progression.

- **Goal:** Quantitatively summarize spatial features in the marked point pattern of Tumor Microenvironment(TME) and investigate their clinical meaning through survival analysis.

- **Features:** 13 spatial functional summary statistics and 2 Areal data indexes.

■ ConvPath pipeline:



METHOD

■ Input data

- (x_i, y_i) : coordinate of cell i in Euclidean space $\mathbb{R}^2$
 $x_i \in [0, 1], y_i \in [0, 1]$ in normalized image.
- i_T, i_I, i_S : different cell types of points, Tumor, Immune, Stromal

■ An example of Spatial summary statistic

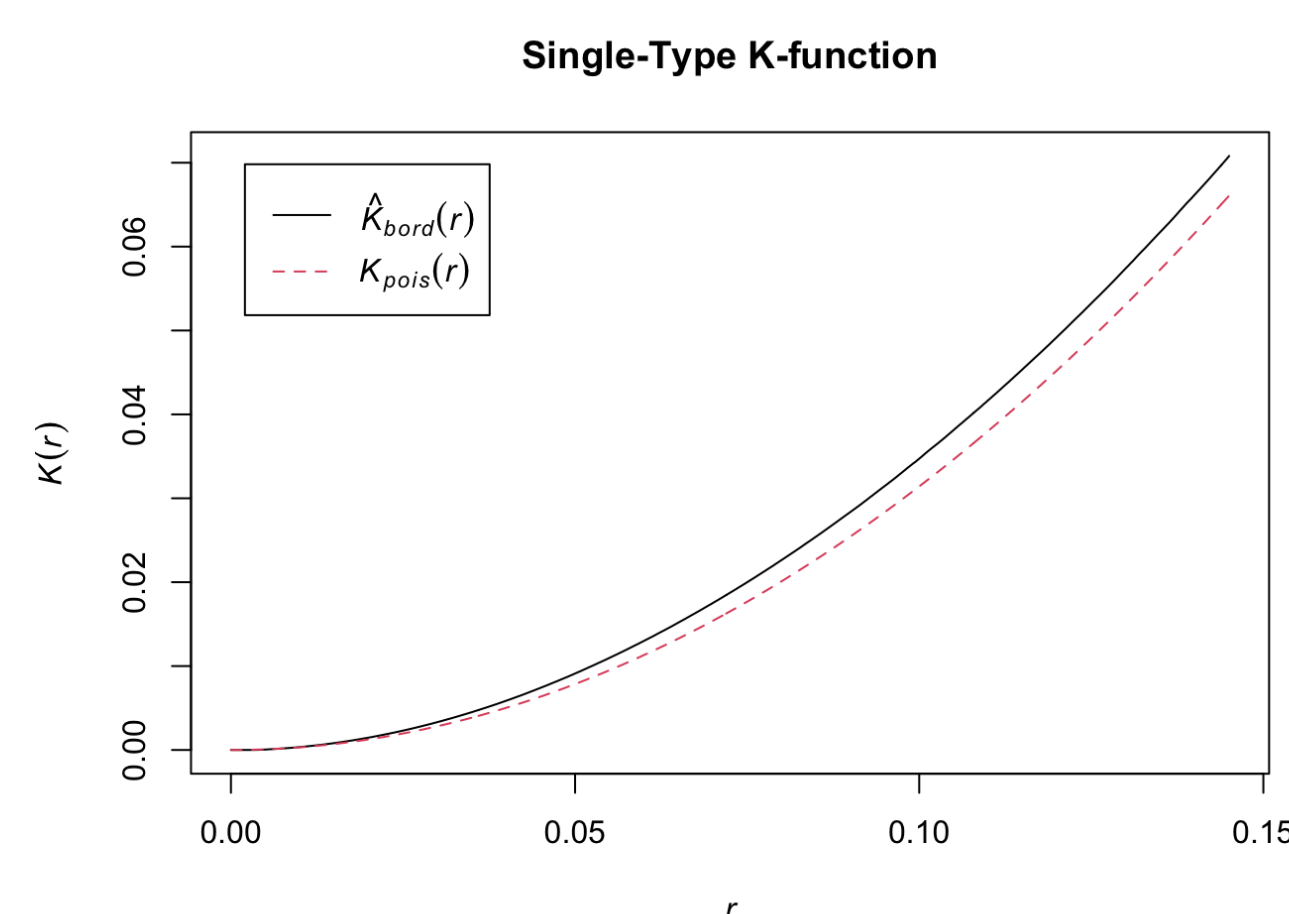
- K-function

$$K(r) = \frac{1}{\lambda^2} \sum_j \sum_{i \neq j} I_r(d_{ij}) w_i, \quad \lambda = \frac{N}{A} \quad \begin{matrix} N : \text{number of points} \\ A : \text{area of the window} \end{matrix}$$

Where:

$$d_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \quad I_r(d_{ij}) = \begin{cases} 1, & \text{if } d_{ij} \leq r \\ 0, & \text{otherwise} \end{cases}$$

$w_i = \frac{1}{e_i}$, e_i represents the proportion of the circle within the window



$K(r) > \pi r^2$ indicates clustered point process

$K(r) < \pi r^2$ indicates inhibited point process

DATA

■ Dataset

- NLST dataset: 1585 image slides, 188 patients
- OPMD dataset: 702 image slides, 128 patients

- - Single-type: spatial correlation within **T, I, or S**
- Cross-type: spatial correlation between **T2I, T2S, S2I**
- Scalar predictors: **Age, Stage**
- **Total of 54 predictors**

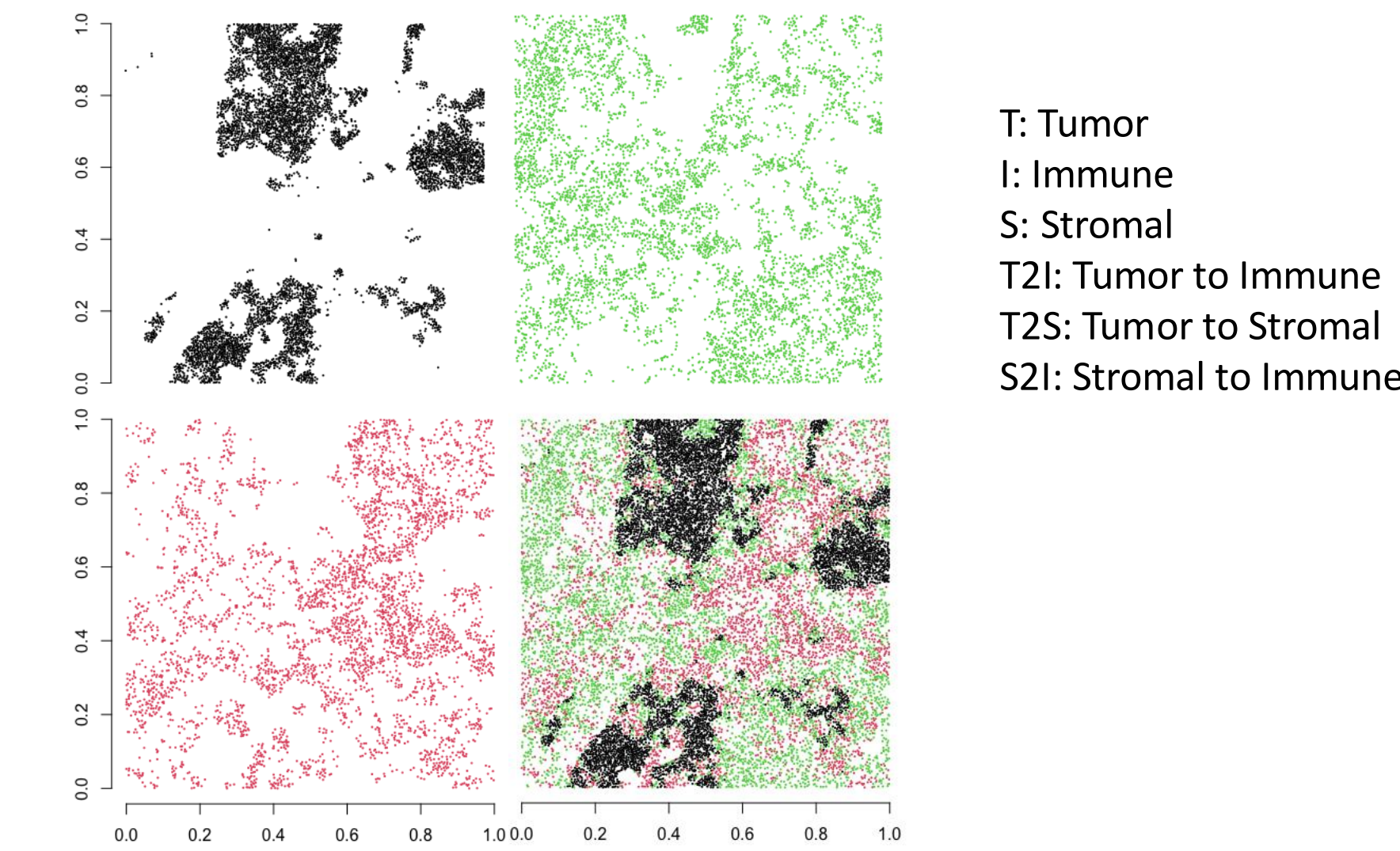


Figure1: Example slide of patient "AA0001"

■ Workflow

- Extracted spatial features per image
- Applied FPCA for functional dimension reduction
- Top 2 FPC scores used as covariates in FCoxPH

■ Functional CoxPH (FCoxPH) model

- To utilize functional data as scalar predictor of CoxPH model.

$$h(t) = h_0(t) \exp \left(Z^T \gamma + \int X^p(u) \alpha(u) du \right)$$

Where:

n-dimensional scalar predictors: $Z = (z_1, \dots, z_n)^T$

functional predictors: $X^p, p = (1, \dots, k)$

baseline hazard function: $h(t), t \in [0, \tau], 0 < \tau < \infty$

■ Karhunen-Loeve Expansion

$$X_i^p(u) = \mu^p(u) + \sum_{j=1}^{\infty} \xi_{ij} \phi_j(u)$$

- FPCA

- FPC Scores are calculated by 2 major PCs that explains $\approx 90\%$ of variance of functional data of image.

- Notations:

Functional Predictor: $X_i^p(u)$

Mean Function: $\mu^p(u) = \mathbb{E}[X_i^p(u)]$

Covariance Decomposition: $\text{Cov}(X^p(u), X^p(u')) = \sum_{j=1}^{\infty} \lambda_j \phi_j(u) \phi_j(u')$

FPCA Scores: $\xi_{ij} = \int (X_i^p(u) - \mu^p(u)) \phi_j(u) du$

Truncated Representation: $X_i^p(u) \approx \mu^p(u) + \sum_{j=1}^q \xi_{ij} \phi_j(u)$

FCoxPH Model: $h_i(t) \approx h_0(t) \exp \left(Z_i^T \gamma + \sum_{j=1}^q \xi_{ij} \alpha_j \right)$

RESULTS

Survival Analysis

■ Multivariate FCoxPH model

- Adjusted for other scalar variables such as age, tumor stage.

■ Single-type J-function

- J-function captures the balance between how likely a random location is near a point and how likely points are near each other.

Predictor	Coef (NLST)	Coef (OPMD)	P-value (NLST)	P-value (OPMD)
J.REP.PC1	-5.337e-18	-3.785e-16	0.384	0.398
J.REP.PC2	3.404e-17	-2.870e-15	0.158	0.067
J.REP.I.PC1	1.370e-17	-1.176e-17	0.104	0.997
J.REP.I.PC2	3.634e-17	-9.803e-15	0.299	0.153
J.REP.T.PC1	1.949e-18	2.405e-16	0.370	0.542
J.REP.T.PC2	2.343e-17	-3.785e-15	0.557	0.020
J.REP.S.PC1	-1.532e-21	5.269e-16	0.993	0.420
J.REP.S.PC2	-1.967e-19	-2.637e-15	0.248	0.083

Table1: Summary of J-function predictors

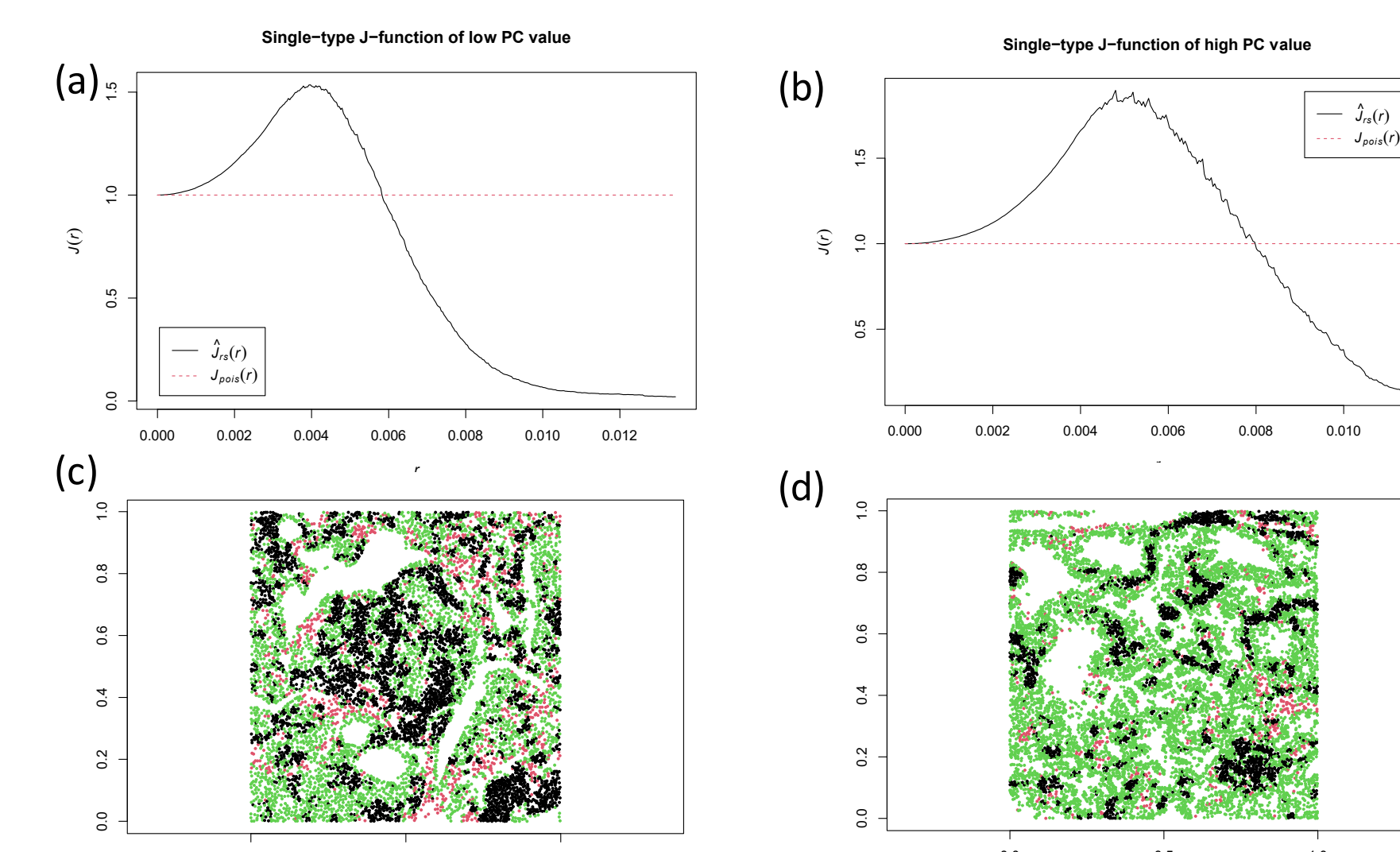


Figure2: (a) J-function of low-risk group image, (b) J-function of high-risk group image, (c) low-risk group image, (d) high-risk group image

■ Cross-type Pair correlation function

- pair correlation function captures the relative likelihood of finding a point at an exact distance r from another point.

Predictor	Coef (NLST)	Coef (OPMD)	P-value (NLST)	P-value (OPMD)
PCF.CROSS.T2I.PC1	1.711e-17	-7.463e-18	0.028	0.742
PCF.CROSS.T2I.PC2	-1.541e-17	-6.748e-16	0.362	0.142
PCF.CROSS.T2S.PC1	-1.170e-18	-1.407e-17	0.495	0.618
PCF.CROSS.T2S.PC2	-1.885e-17	-3.019e-16	0.683	0.305
PCF.CROSS.S2I.PC1	-3.951e-18	-1.309e-17	0.723	0.565
PCF.CROSS.S2I.PC2	8.359e-18	-4.462e-16	0.700	0.165

Table2: Summary of Pair correlation function predictors

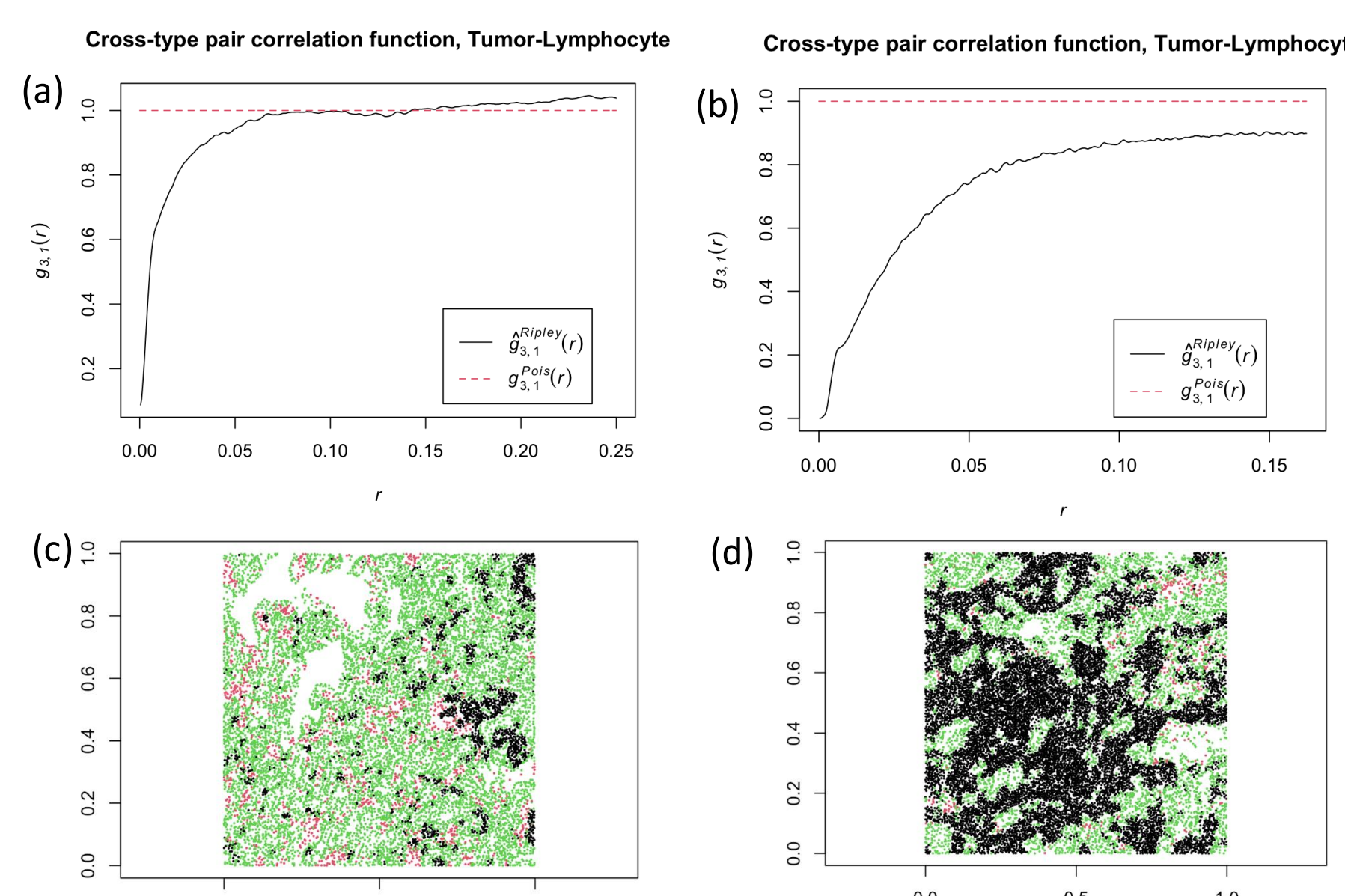


Figure3: (a) Pair correlation function of low-risk group image, (b) Pair correlation function of high-risk group image, (c) low-risk group image, (d) high-risk group image

*Bolding signifies features with p-value ≤ 0.20

RESULTS

■ Cross-type G-function

- G-function captures the cumulative distribution of nearest-neighbor distances between points.

Predictor	Coef (NLST)	Coef (OPMD)	P-value (NLST)	P-value (OPMD)
G.CROSS.T2I.PC1	-2.292e-17	-4.081e-18	0.109	0.274
G.CROSS.T2I.PC2	-5.330e-17	-3.174e-16	0.587	0.117

Table3: Summary of G-function predictors

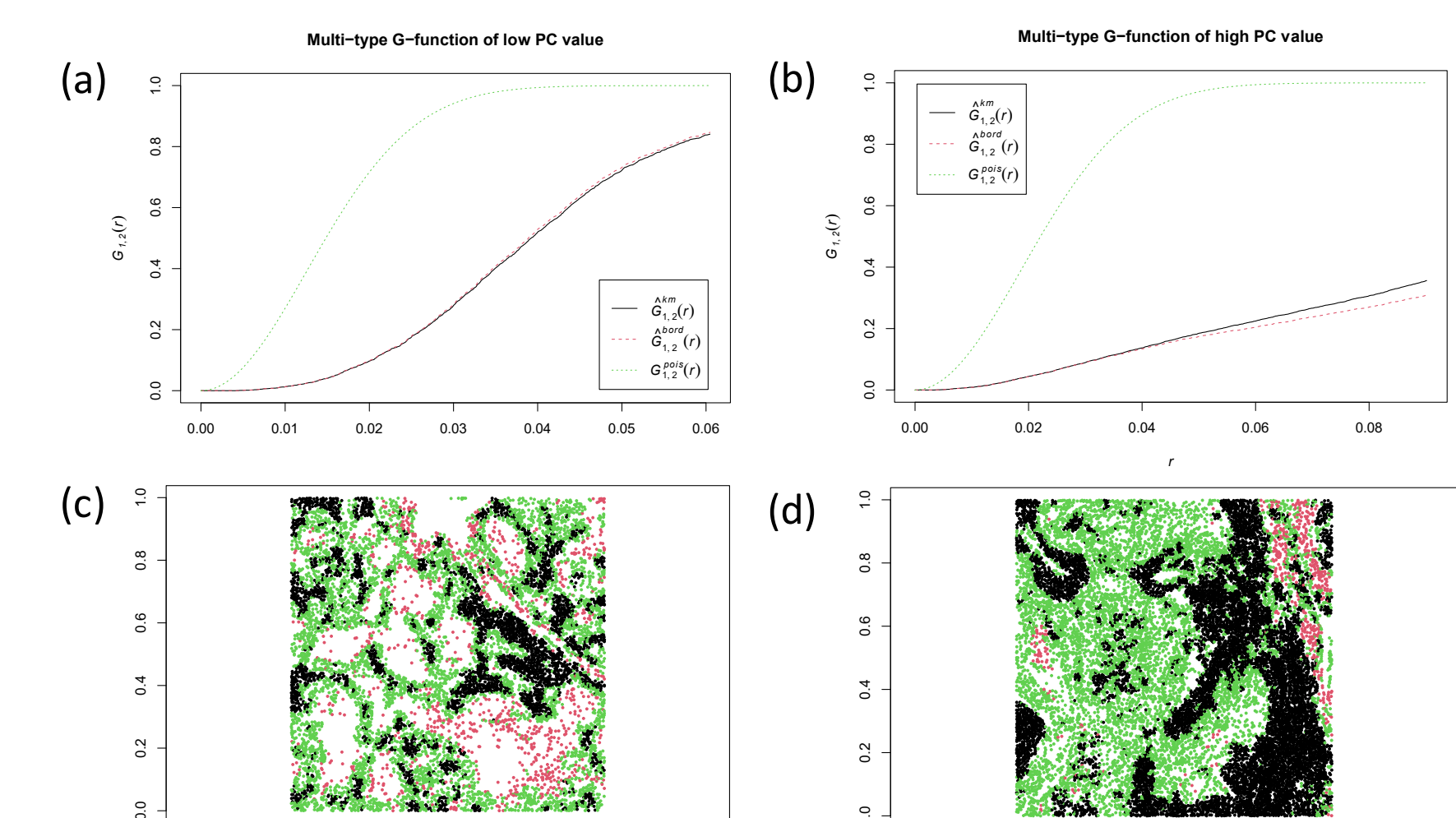


Figure4: (a) G-function of low-risk group image, (b) G-function of high-risk group image, (c) low-risk group image, (d) high-risk group image

Downstream Analysis

■ LOOCV(Leave One Out Cross Validation)

- Trained (N-1) images to predict the risk score of each image
- Averaged risk score of each patient from all associated images.

■ Kaplan-Meier Survival curves

- Grouped patient into High-risk and Low-risk groups
- Log-rank test showed significance difference between groups.

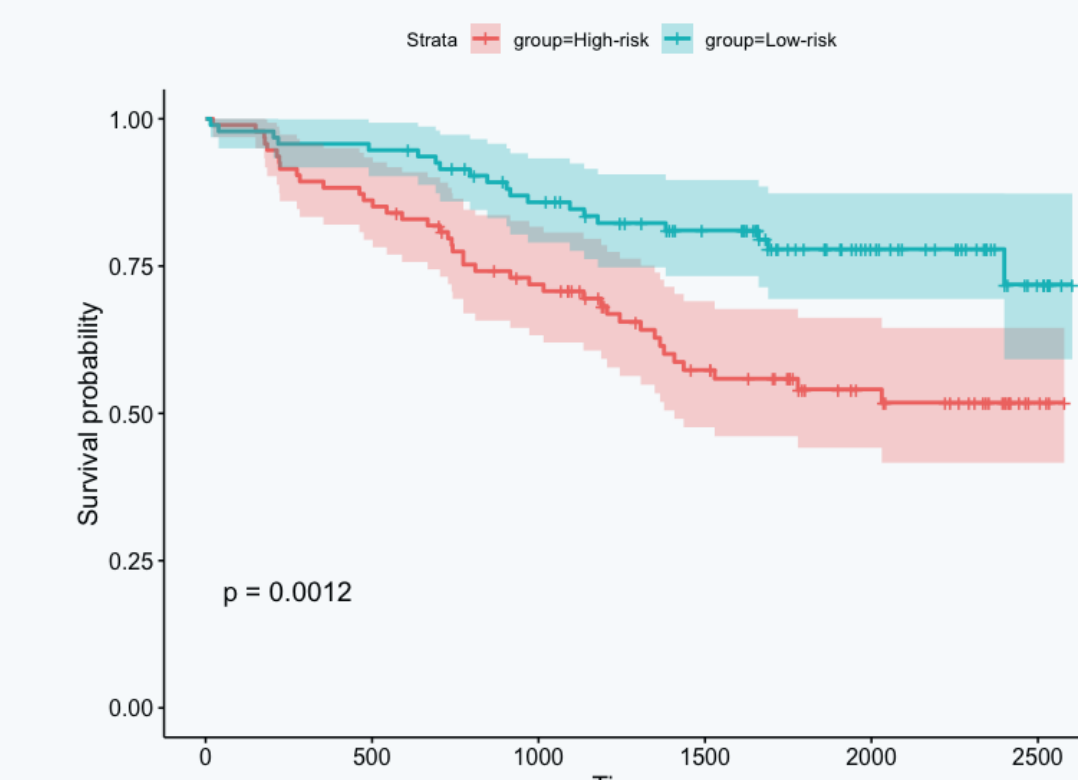


Figure5: Kaplan-Meier plot between high-risk, low-risk groups

Log-rank Test Summary				
Group	N	Observed	Expected	$(O - E)^2 / V$
High-Risk	94	40	27.5	10.5
Low-Risk	94	20	32.5	10.5

Table4: Log-rank test

SUMMARY

■ Spatial correlation in Tumor Microenvironment

- Statistically significant spatial features were extracted from pathology slides. Among 54 candidate predictors, **12 (NLST)** and **9 (OPMD)** variables showed statistically significant spatial associations.

REFERENCE

Li X. Deciphering cell to cell spatial relationship for pathology images using SpatialQPFs. Sci Rep. 2024 Nov 28;14(1):29585. doi: 10.1038/s41598-024-81383-1. PMID: 39609630. PMCID: PMC11605059

Chul Moon, Qiwei Li, Guanghua Xiao "Using persistent homology topological features to characterize medical images: Case studies on lung and brain cancers," The Annals of Applied Statistics, Ann. Appl. Stat. 17(3), 2192-2211, (September 2023)